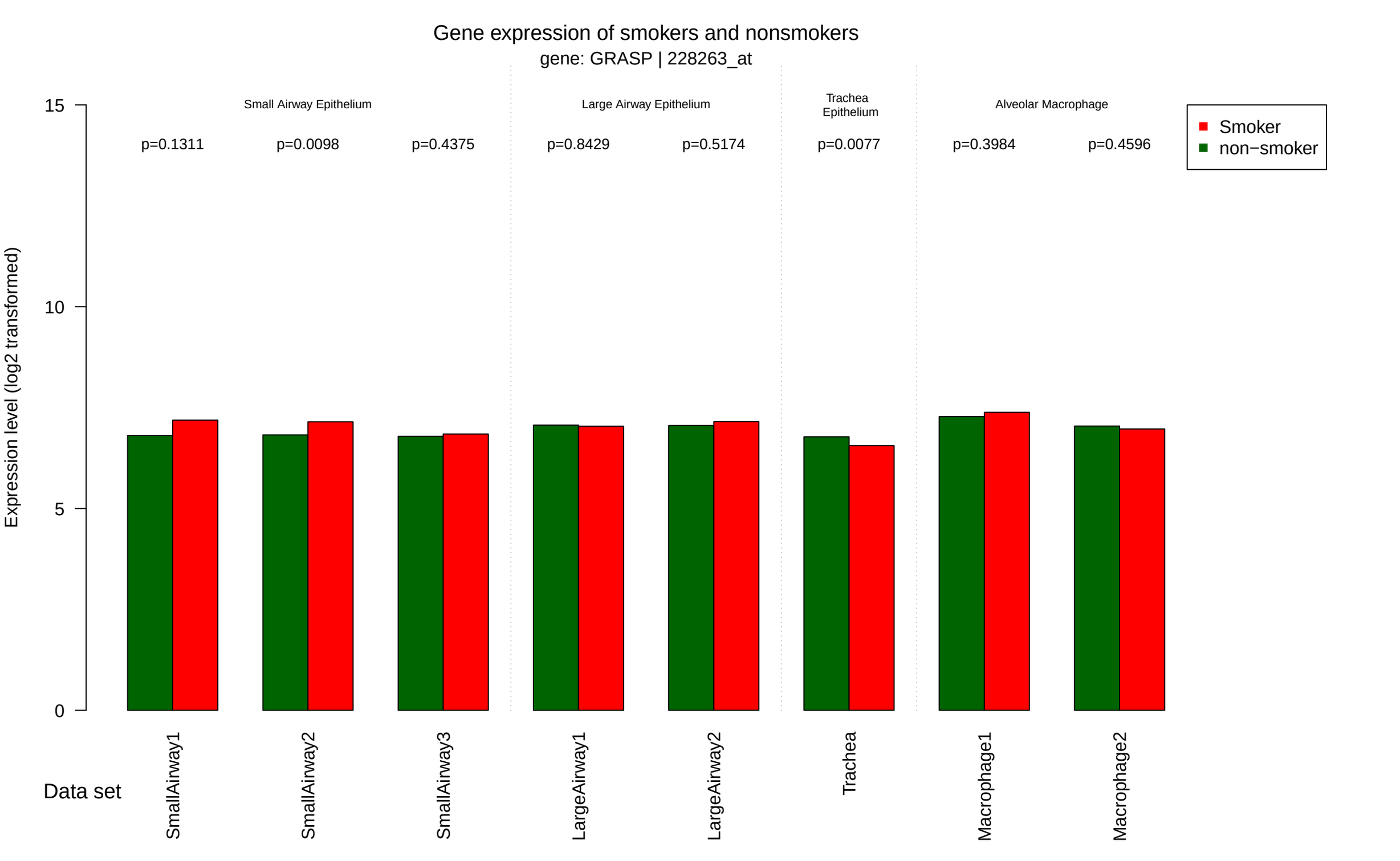
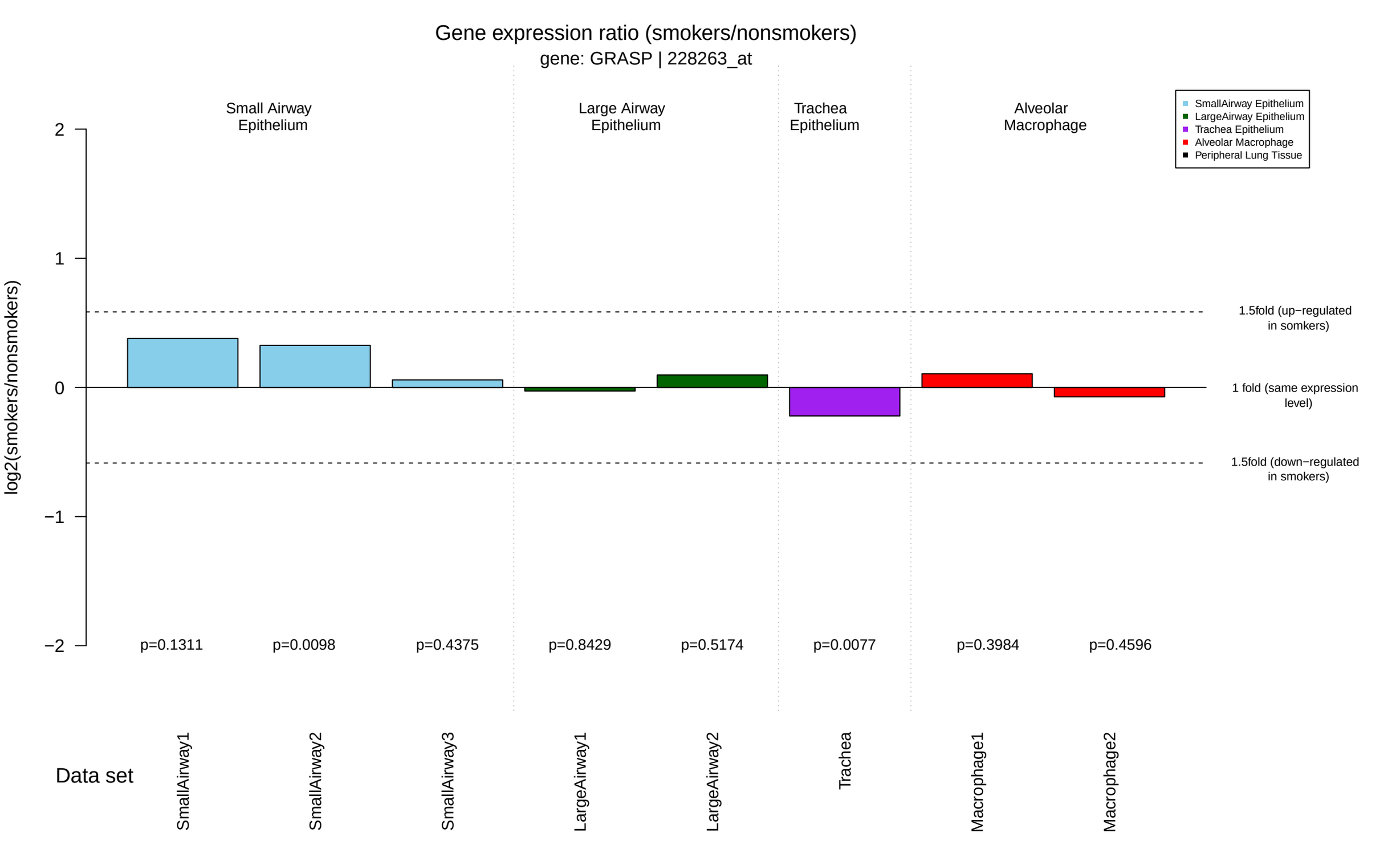
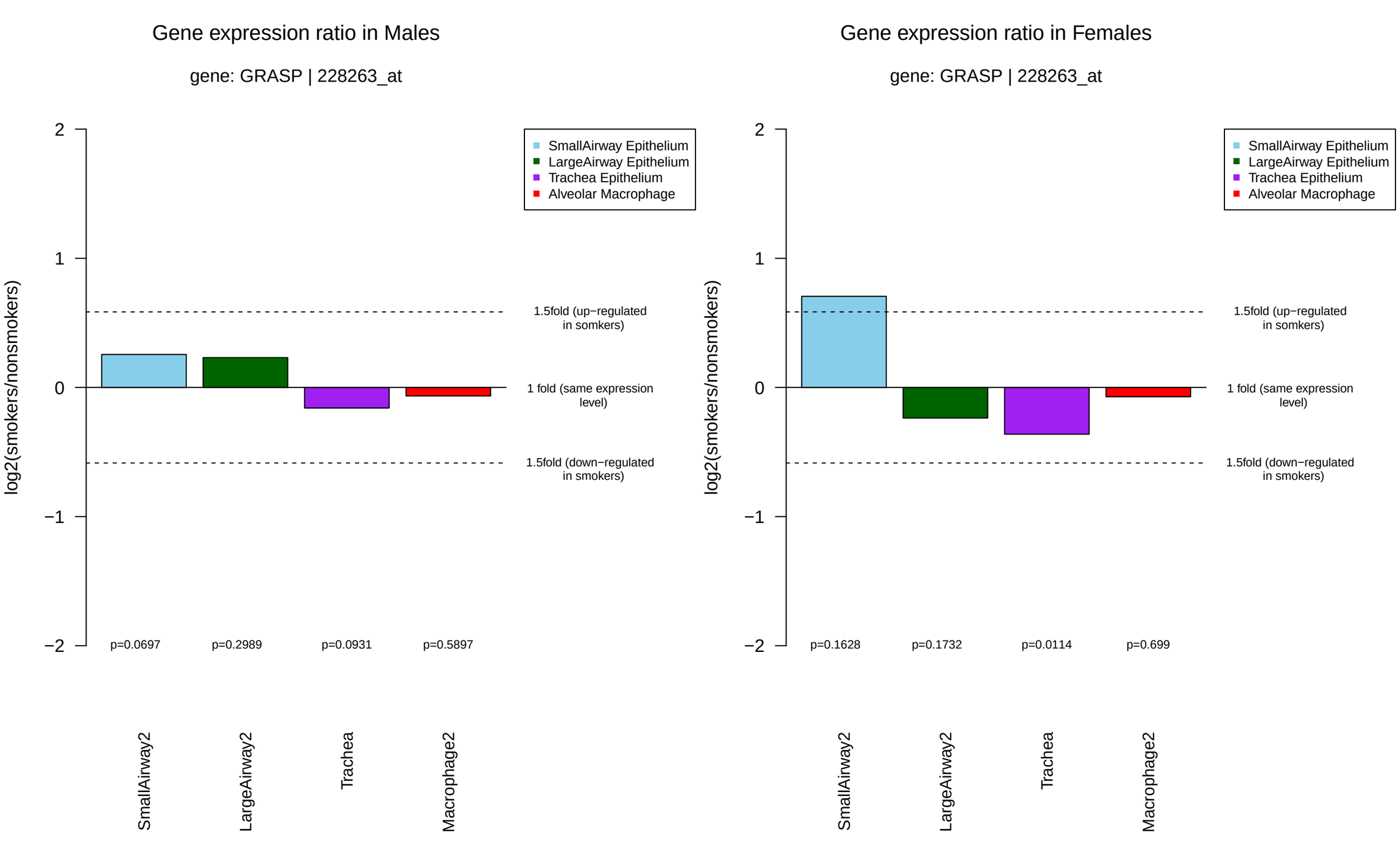
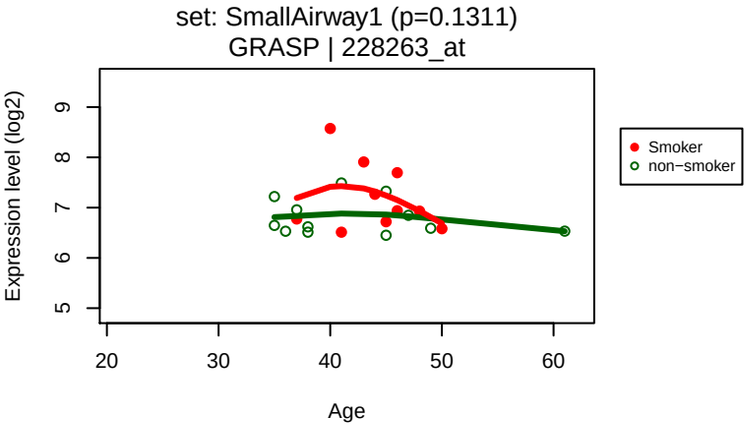
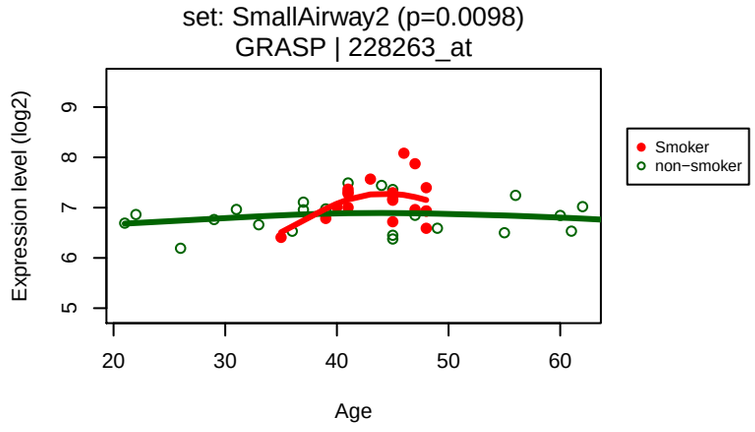
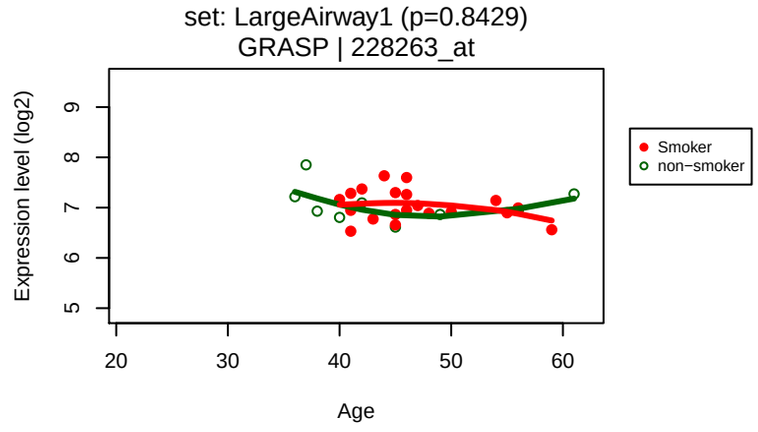

Gene expression of smokers and nonsmokers
gene: GRASP | 228263_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1311
p=0.0098
p=0.4375
p=0.8429
p=0.5174
p=0.0077
p=0.3984
p=0.4596
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

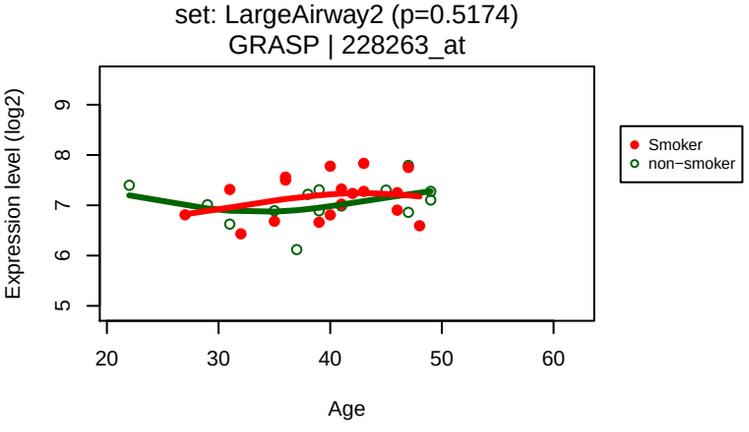

Gene expression ratio (smokers/nonsmokers)
gene: GRASP | 228263_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1311
p=0.0098
p=0.4375
p=0.8429
p=0.5174
p=0.0077
p=0.3984
p=0.4596
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

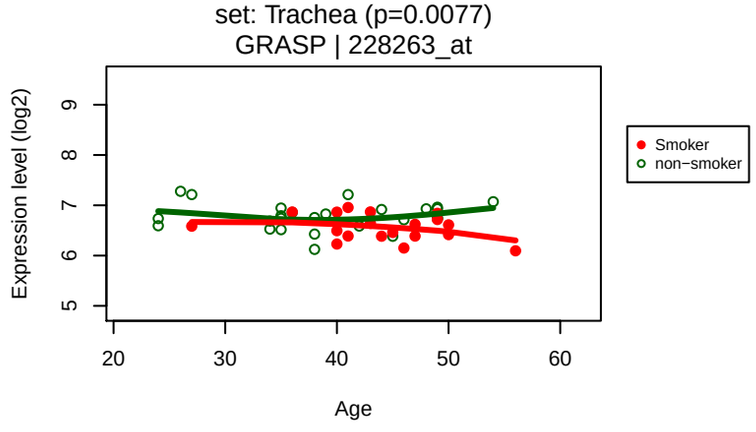

Gene expression ratio in Males
gene: GRASP | 228263_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0697
p=0.2989
p=0.0931
p=0.5897
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRASP | 228263_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1628
p=0.1732
p=0.0114
p=0.699
SmallAirway2
LargeAirway2
Trachea
Macrophage2

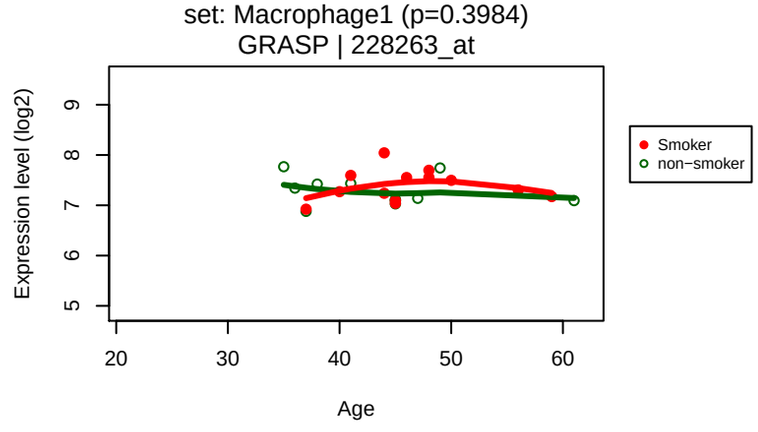

set: SmallAirway1 (p=0.1311)
GRASP | 228263_at
Expression level (log2)
Age
Smoker
non−smoker

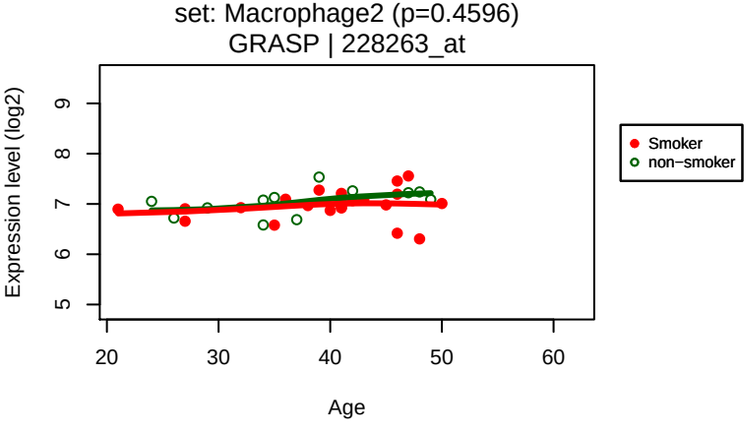

set: SmallAirway2 (p=0.0098)
GRASP | 228263_at
Expression level (log2)
Age
Smoker
non−smoker

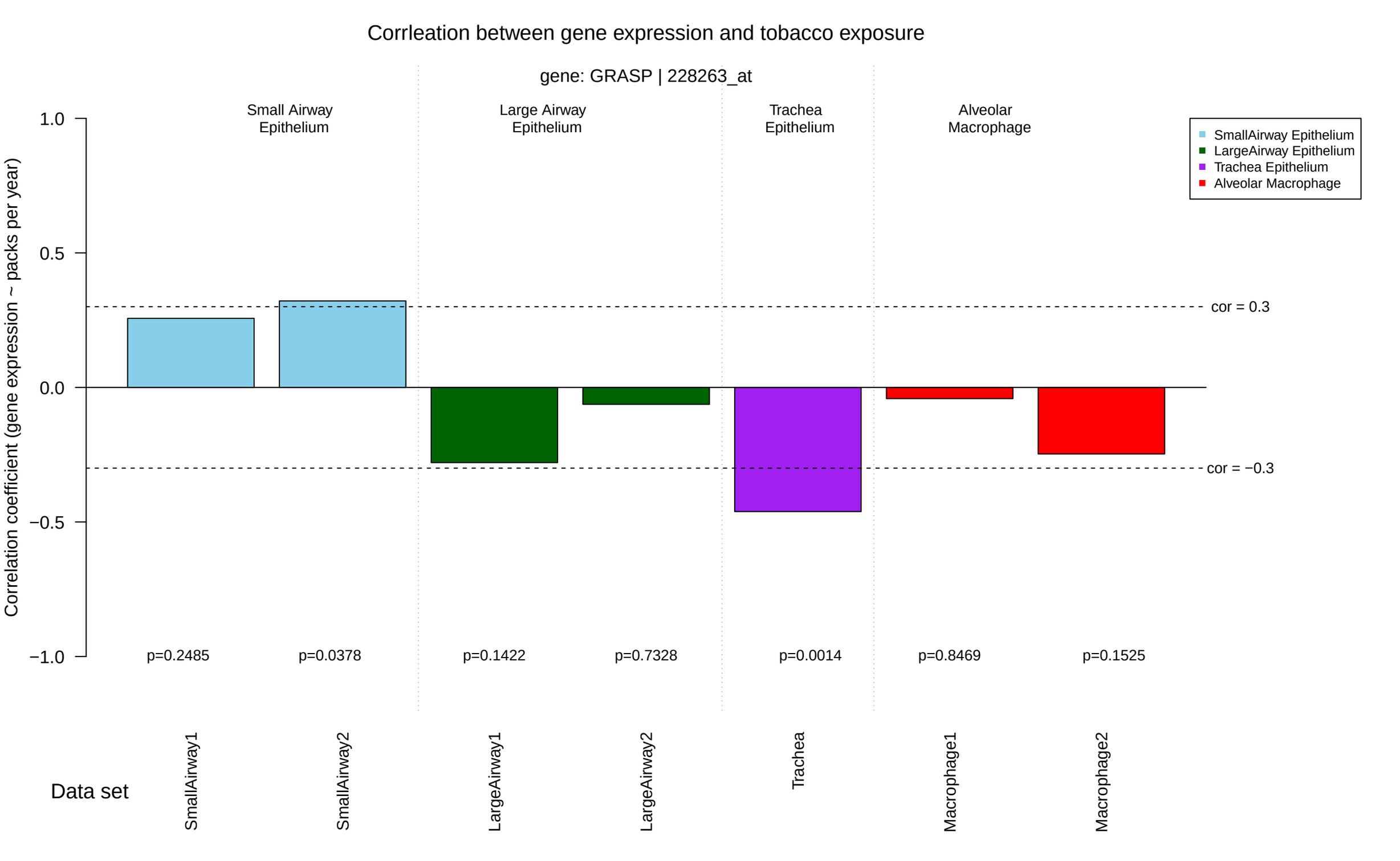

set: LargeAirway1 (p=0.8429)
GRASP | 228263_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5174)
GRASP | 228263_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0077)
GRASP | 228263_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3984)
GRASP | 228263_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4596)
GRASP | 228263_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRASP | 228263_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2485
p=0.0378
p=0.1422
p=0.7328
p=0.0014
p=0.8469
p=0.1525
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2